THE QUOTIENT-STACK SQUARE IS A 2-FIBRE PRODUCT OF STACKS IN GROUPOIDS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original lemma 044R: sources/stacks-chow/tags.txt, source lines 5410–5410. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 044S: sources/stacks-chow/tags.txt, source lines 5411–5411. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02ZL: sources/stacks-chow/tags.txt, source lines 3916–3916. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0040: sources/stacks-chow/tags.txt, source lines 156–156. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 044V: sources/stacks-chow/tags.txt, source lines 5414–5414. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04Y4: sources/stacks-chow/tags.txt, source lines 6417–6417. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Assume $B \to S$, (U, R, s, t, c), and $\pi : \mathcal{S}_U \to [U/R]$ are as in Lemma 044R. The 2-commutative square*

$$\begin{array}{ccc} \mathcal{S}_R & \xrightarrow[s]{} & \mathcal{S}_U \\ {\scriptstyle t}\downarrow & & \downarrow{\scriptstyle \pi} \\ \mathcal{S}_U & \xrightarrow{\pi} & [U/R] \end{array}$$

of Lemma 044S is a 2-fibre product of stacks in groupoids of $(Sch/S)_{fppf}$.

Proof. According to Stacks, Lemma 02ZL the lemma makes sense. It also tells us that we have to show that the functor

$$\mathcal{S}_R \longrightarrow \mathcal{S}_U \times_{[U/R]} \mathcal{S}_U$$

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which maps $r : T \to R$ to $(T, t(r), s(r), \alpha(r))$ is an equivalence, where the right hand side is the 2-fibre product as described in Categories, Lemma 0040. This is, after spelling out the definitions, exactly the content of Lemma 044V. (Alternative proof: Work out the meaning of Lemma 04Y4 in this situation will give you the result also.) $\square$